THE VALUE, WITH ITS EVIDENCE.

Sample engine report

01 Executive summary

3 years · USD · discount rate 10.0%

Measure	Hard savings	Capacity-inclusive
Net present value	-$18,641.93	$127,956.73
ROI · undiscounted	-15.6%	202.9%
ROI · NPV / PV of costs	-24.8%	169.9%
Payback	No payback within the selected horizon	about 11 months (uniform-flow assumption)
Benefits · present value	All included drivers	$203,275.29
Costs · present value	All cost lines	$75,318.56

Confidence 94%, based on 7 documented inputs.

7 documented inputs, 0 measured. Effective evidence discounts: time savings 6.5%, data quality 4.5%, risk reduction 6.5%. Minimum confidence required: 85.0%. Capped by Time savings.

Over three years this project is estimated to return a capacity-inclusive NPV of $127,957 on costs with a present value of $75,319 [D1, D2, D3]. Hard savings alone give an NPV of $-18,642 [B3]. All inputs are documented rather than measured, so treat this as a pre-funding estimate.

STUB · sample assessment data

This report renders the stored engine output for the recorded run. The exporter formats supplied values and does not recompute financial metrics. Refer to the recorded evidence before making a funding decision.

02 Context and assumptions

Run run_sample_released. Expected scenario. Generated 2026-11-05T12:00:00Z. Contract 1; formula 0.1; settings s1.

Input snapshot not supplied

The stored step input arrays are empty. Source formulas and source types are preserved below; missing entered inputs and project context have not been invented.

Accepted proxy	Value	Source / evidence level
Freed time used productively (%)	50	Nucleus productivity correction factor 0.5 · benchmark
Discount rate (%)	10	Forrester TEI convention (organizations typically 8-16%) · benchmark

03 Benefits by driver

Adjusted values over the selected horizon

Driver / class	Evidence discount	Confidence	Total	PV
Time savings · Capacity value · included	6.5%	93.5%	$102,850.00	$83,770.66
Data quality · Hard savings · included	4.5%	95.5%	$69,585.12	$56,676.63
Risk reduction · Expected avoided loss · included	6.5%	93.5%	$77,137.50	$62,828.00

Driver contribution · horizon totals

03

Time savings · show the math

Capacity value · evidence discount 6.5% · confidence 93.5%

0 measured · 1 documented · 0 benchmark · 0 estimate · 0 assumption

Ref	Step / formula / source	Year 1	Year 2	Year 3	Horizon total
A1	1. Gross annual value staff x hours saved per week x weeks x loaded rate (10 x 5 x 40 x $50) Source: user; evidence level: documented	$100,000.00	$100,000.00	$100,000.00	$300,000.00
A2	2. After adoption (50% / 80% / 90%) gross x adoption by year Source: calculated; evidence level: calculated	$50,000.00	$80,000.00	$90,000.00	$220,000.00
A3	3. After productive use of freed time (capacity value only) after adoption x productive use of freed time 50% Source: default; evidence level: benchmark	$25,000.00	$40,000.00	$45,000.00	$110,000.00
A4	4. After evidence discount (6.5%) previous step x (1 - evidence discount) Source: calculated; evidence level: calculated	$23,375.00	$37,400.00	$42,075.00	$102,850.00

What makes this vary

Freed time may not be turned into other work (productive use of freed time is a default of 50%) Adoption ramp depends on training

How measured: Time study on the support queue before and after launch, reviewed monthly by the ops lead

Freed time is capacity value, not cash [A4].

03

Data quality · show the math

Hard savings · evidence discount 4.5% · confidence 95.5%

0 measured · 3 documented · 0 benchmark · 0 estimate · 0 assumption

Ref	Step / formula / source	Year 1	Year 2	Year 3	Horizon total
B1	1. Gross annual value (errors before - errors after) x cost per error ((72 - 24) x $690) Source: user; evidence level: documented	$33,120.00	$33,120.00	$33,120.00	$99,360.00
B2	2. After adoption (50% / 80% / 90%) gross x adoption by year Source: calculated; evidence level: calculated	$16,560.00	$26,496.00	$29,808.00	$72,864.00
B3	3. After evidence discount (4.5%) previous step x (1 - evidence discount) Source: calculated; evidence level: calculated	$15,814.80	$25,303.68	$28,466.64	$69,585.12

What makes this vary

Error baseline covers one year only Cost per error includes only expedite freight

How measured: Monthly expedite-freight invoices compared with the 12-month baseline

Fewer expedited shipments are hard savings [B3].

03

Risk reduction · show the math

Expected avoided loss · evidence discount 6.5% · confidence 93.5%

0 measured · 3 documented · 0 benchmark · 0 estimate · 0 assumption

Ref	Step / formula / source	Year 1	Year 2	Year 3	Horizon total
C1	1. Gross annual value (ALE before - ALE after), ALE = events per year x loss per event ((3 - 1.5) x $25,000) Source: user; evidence level: documented	$37,500.00	$37,500.00	$37,500.00	$112,500.00
C2	2. After adoption (50% / 80% / 90%) gross x adoption by year Source: calculated; evidence level: calculated	$18,750.00	$30,000.00	$33,750.00	$82,500.00
C3	3. After evidence discount (6.5%) previous step x (1 - evidence discount) Source: calculated; evidence level: calculated	$17,531.25	$28,050.00	$31,556.25	$77,137.50

What makes this vary

Outage frequency is a small sample Loss per event excludes reputational cost

How measured: Incident log: sustained outages per year, reviewed quarterly

Halving outage frequency removes expected loss [C3].

04 Costs and uplift

PV $75,318.56 · delay 0 months

Ref / cost	Basis	Base	Uplift	Adjusted
D1 · Vendor license, integration and migration (one-time)	quoted	$30,000.00	15.0%	$34,500.00
D2 · Training and change management (one-time)	estimated	$5,000.00	30.0%	$6,500.00
D3 · Annual run cost	quoted	$12,000.00	15.0%	$13,800.00

05 Unquantified benefits and flexibility

Benefit	Reason	What to add
Decision speed	Not selected as a goal for this project	Add the decision speed goal and a dated example decision

The data pipeline built here could be reused for a second team at lower cost.

06

Financial summary

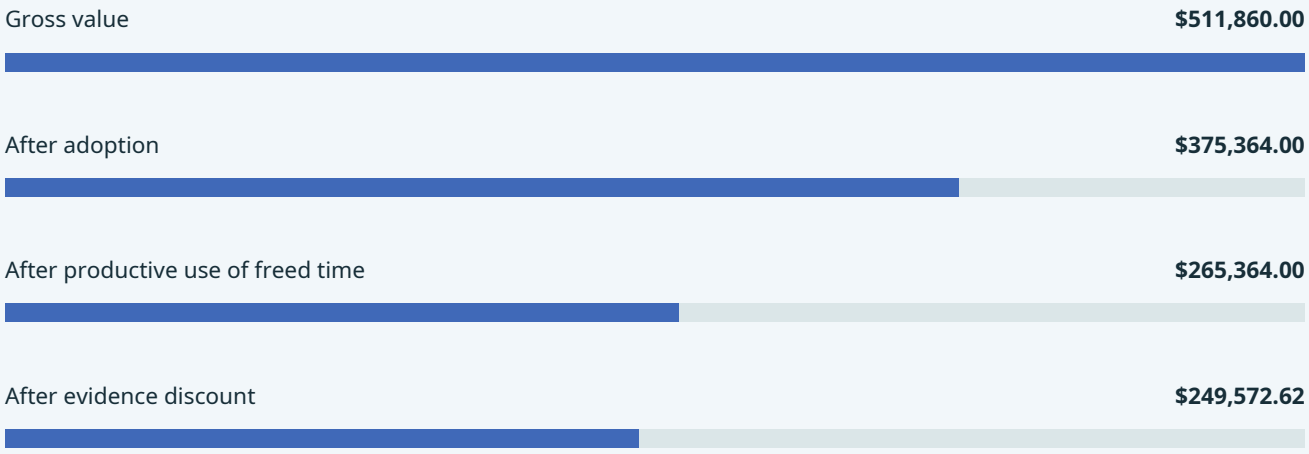
Undiscounted flows; net and cumulative are capacity-inclusive.

Period	Hard benefit	Capacity / avoided loss	Cost	Net inclusive	Cumulative inclusive
Time zero	$0.00	$0.00	$41,000.00	-$41,000.00	-$41,000.00
Year 1	$15,814.80	$40,906.25	$13,800.00	$42,921.05	$1,921.05
Year 2	$25,303.68	$65,450.00	$13,800.00	$76,953.68	$78,874.73
Year 3	$28,466.64	$73,631.25	$13,800.00	$88,297.89	$167,172.62

Financial basis

Capacity value and expected avoided loss are distinct from documented hard savings. The net and cumulative figures include these benefits; they do not establish a cash reduction.

Adjustment stages · supplied horizon levels

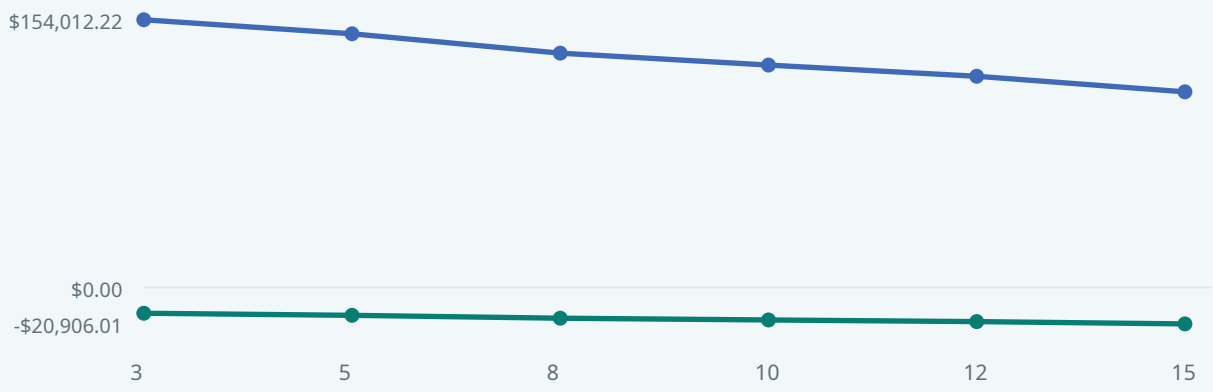


07 Scenarios and sensitivity

Scenario / assumptions	NPV hard	NPV inclusive	ROI · PV-based inclusive
conservative: Evidence discounts at the top of each band, adoption x0.85, productive use of freed time 40%	-$29,917.91	$76,318.52	101.3%
expected: Headline scenario: stated inputs and computed evidence discounts	-$18,641.93	$127,956.73	169.9%
optimistic: Evidence discounts at the bottom of each band, adoption x1.10 (capped at 99%), productive use of freed time 60%	-$10,036.57	$182,143.18	241.8%

NPV sensitivity · discount rate

Teal: hard savings · blue: capacity-inclusive



Discount rate (%)	NPV hard	NPV capacity-inclusive
3	-$14,778.51	$154,012.22
5	-$15,977.37	$145,951.48
8	-$17,629.04	$134,810.80
10	-$18,641.93	$127,956.73
12	-$19,590.98	$121,518.06
15	-$20,906.01	$112,567.99

08 Risks of doing nothing vs proceeding

Decision	Risks
Doing nothing	Expedite-freight spend and outage exposure continue at today's rates.
Proceeding	Adoption may ramp more slowly than 50% / 80% / 90%.
Proceeding	Vendor costs are quoted, but internal training time is estimated.

09 Measurement plan

Driver / metric	Baseline / source	Owner	Frequency
Time savings: Time study before and after launch	30 minutes per ticket; Ops time sample, Q3	Ops lead	monthly
Data quality: Expedite invoices vs 12-month baseline	72 expedited shipments per year; Finance export	Finance analyst	monthly
Risk reduction: Sustained outages per year	3 per year; Incident log	IT manager	quarterly

10 Appendix and audit trail

Field	Recorded value
Run	run_sample_released
Mode	stub
Contract	1
Formula	0.1
Settings	s1
Generated	2026-11-05T12:00:00Z
Evidence evidence levels	0 measured · 7 documented · 0 benchmark · 0 estimate · 0 assumption
Narrative references	References such as A4, B3 and D1–D3 refer to the step and cost tables in this report.

Engine integrity check	Status
bounds	Passed
versions recorded	Passed
every number traces to a step	Passed
driver totals reconcile to benefit total	Passed
cash flows reproduce NPV	Passed
no unresolved double count	Passed
no missing essential cost	Passed

No engine warnings were recorded.

No overrides were recorded.